

# Gaussian elimination

## How to solve the standard problems of Lectures 0–5

Most computational problems of the first lectures are solved by one algorithm: write a matrix, reduce it to RREF, and read the answer off the pivots. The problems differ only in *which* matrix you write and *what* you read off. The elimination steps are written out once, in Lecture 0; here they are marked by a single  $\xrightarrow{\text{RREF}}$  arrow. Each method ends with the sheet problems it solves.

### Choosing the matrix

Vectors go into the *columns*; equations go into the *rows*. A bar appears only when the question names a target. Vectors of any other space are first replaced by their coordinate vectors in one fixed basis; for polynomials and matrices, the standard one (Example 2.15).

| Lect. | Question                                                                    | Matrix                                              | Bar                              | Read-off                                                                                               |
|-------|-----------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------|
| 0     | Solve $Ax = b$ ; the null space                                             | equations as rows                                   | $  b$                            | a row $[0 \cdots 0   c]$ , $c \neq 0$ : none; else $x_p$ plus one vector per free variable (Section 1) |
| 0     | The inverse $A^{-1}$                                                        | $A$                                                 | $  I$                            | left half $I$ : the right half; else no inverse (Section 1)                                            |
| 2     | $v \in \text{span}(v_1, \dots, v_m)$ ?<br>Coordinates $[v]_{\mathcal{B}}$ ? | the vectors as columns                              | $  v$                            | consistent? the solution (Section 2)                                                                   |
| 2     | Independent? A basis of the span?                                           | the vectors as columns, in order                    | —                                | the pivot columns (Section 3)                                                                          |
| 2     | $U + W$ and $U \cap W$                                                      | bases of $U$ , then $W$ , as columns                | —                                | pivots; relations (Section 4)                                                                          |
| 3     | $\ker T$ and $\text{im } T$ ,<br>$Tx = Ax$                                  | $A$                                                 | —                                | free variables; pivot columns of $A$ (Section 5)                                                       |
| 4     | Matrix of $T$ , basis $C$ of the codomain                                   | $C$ as columns                                      | $  Tv_1, \dots, Tv_n$            | the right half (Section 6)                                                                             |
| 4     | Change of basis: $[v]_{\mathcal{B}'}$ , $A'$                                | $P$                                                 | $  [v]_{\mathcal{B}};$<br>$  AP$ | the right half (Section 6)                                                                             |
| 5     | Eigenvalues and eigenspaces                                                 | $A - \lambda I$ for each root $\lambda$ of $\chi_A$ | —                                | free variables: a basis of $E(\lambda, A)$ (Section 7)                                                 |

## 1 Linear systems

### Lecture 0

**Problem.** Solve  $Ax = b$ : decide whether it has solutions, and write them all. For  $b = 0$  this asks for a basis of the null space  $\ker A$ .

**Method**

1. Reduce  $[A \mid b]$  to RREF. A row  $[0 \cdots 0 \mid c]$  with  $c \neq 0$ : no solution.
2. Set every free variable to 0 and read off a particular solution  $x_p$ .
3. Take  $b = 0$ . For each free variable, set it to 1 and the other free variables to 0, and read off one vector. These vectors are a basis of  $\ker A$ .
4. The solution set is  $x_p$  plus any linear combination of the basis of  $\ker A$ .

**Example.** Solve the system  $x_1 - 2x_2 + 2x_3 + x_4 = 2$ ,  $3x_1 - 6x_2 + 5x_3 + 4x_4 = 7$ :

$$\left[ \begin{array}{cccc|c} 1 & -2 & 2 & 1 & 2 \\ 3 & -6 & 5 & 4 & 7 \end{array} \right] \xrightarrow{\text{RREF}} \left[ \begin{array}{cccc|c} \textcircled{1} & -2 & 0 & 3 & 4 \\ 0 & 0 & \textcircled{1} & -1 & -1 \end{array} \right].$$

The free variables are  $x_2$  and  $x_4$ , and the pivot variables are  $x_1 = 2x_2 - 3x_4 + 4$  and  $x_3 = x_4 - 1$ . Then:

$$\begin{aligned} x_2 = 0, x_4 = 0 &\implies x_p = (4, 0, -1, 0), \\ b = 0, x_2 = 1, x_4 = 0 &\implies u_1 = (2, 1, 0, 0), \\ b = 0, x_2 = 0, x_4 = 1 &\implies u_2 = (-3, 0, 1, 1). \end{aligned}$$

The general solution:

$$x = (4, 0, -1, 0) + s(2, 1, 0, 0) + t(-3, 0, 1, 1),$$

where  $s$  and  $t$  are arbitrary numbers. ◀

**Check.** Substitute  $x_p$ ; multiply  $A$  by each null-space vector.

**Variant: the inverse.** Reduce  $[A \mid I]$ . If the left half becomes  $I$ , the right half is  $A^{-1}$ ; a zero row on the left means  $A$  is not invertible.

**Practice.** Sheet 0: 0.22, 0.25, 0.26, 0.30, 0.31; 0.29 (split into cases before dividing by an expression in  $a$ ); 0.23 (null space); 0.24, 0.27 (inverse). Sheet 2: 2.13, 2.16 (the solution set of a homogeneous system).

## 2 Span membership and coordinates

### Lecture 2

**Problem.** Is  $v$  in  $\text{span}(v_1, \dots, v_m)$ , and with which coefficients? Coordinates  $[v]_{\mathcal{B}}$  ask the same for a basis  $\mathcal{B} = (v_1, \dots, v_m)$ .

**Method**

1. Write  $v_1, \dots, v_m$  into the columns and  $v$  after the bar; reduce.
2. A row  $[0 \cdots 0 \mid c]$  with  $c \neq 0$ :  $v$  is not in the span.
3. Otherwise the solution gives the coefficients. For a basis it is unique: it is  $[v]_{\mathcal{B}}$ .

**Example.** Is  $v = (7, 5, 8)$  in  $\text{span}(v_1, v_2)$ , where  $v_1 = (1, -1, 2)$  and  $v_2 = (3, 1, 4)$ ? We need numbers  $a$  and  $b$  such that  $v = av_1 + bv_2$ :

$$\left[ \begin{array}{cc|c} 1 & 3 & 7 \\ -1 & 1 & 5 \\ 2 & 4 & 8 \end{array} \right] \xrightarrow{\text{RREF}} \left[ \begin{array}{cc|c} \textcircled{1} & 0 & -2 \\ 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 \end{array} \right] \implies a = -2, b = 3.$$

The last row is zero, so the system is consistent:

$$v = -2v_1 + 3v_2 \quad \text{and} \quad [v]_{\mathcal{B}} = (-2, 3).$$

Suppose instead that  $v = (7, 5, 9)$ . Then the RREF has the row  $[0 \ 0 \ 1]$ , and  $v \notin \text{span}(v_1, v_2)$ .

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**Practice.** Sheet 2: 2.1–2.4 (span); 2.22–2.25 (coordinates).

### 3 Independence; reducing a list to a basis

Lecture 2

**Problem.** Is  $v_1, \dots, v_m$  independent? If not, which vectors are redundant, and what are a basis and the dimension of the span?

#### Method

1. Write  $v_1, \dots, v_m$  into the columns, in the order of the list; reduce.
2. The list is independent iff every column has a pivot.
3. A column without a pivot is redundant: its entries are its coefficients in the pivot columns to its left.
4. The pivot columns *of the original list* are a basis of the span; the dimension is the number of pivots.

**Why step 4 works.** Deleting the redundant vectors does not change the span (Lemma 2.10). The same row operations take the remaining columns to distinct standard vectors  $e_i$ , so they are independent.

**Example.** Are  $v_1 = (1, 0, 2, 1)$ ,  $v_2 = (2, 1, 3, -1)$ ,  $v_3 = (3, 2, 4, -3)$ ,  $v_4 = (0, 1, 1, 2)$  independent? Find a basis and the dimension of their span.

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 1 \\ 2 & 3 & 4 & 1 \\ 1 & -1 & -3 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & -1 & 0 \\ 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies v_3 = -v_1 + 2v_2.$$

So  $v_1, \dots, v_4$  are dependent. The vectors  $v_1, v_2$  and  $v_4$  are independent and form a basis of  $\text{span}(v_1, v_2, v_3, v_4)$ , of dimension 3.

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**Variants.** Extend an independent  $u_1, \dots, u_k$  to a basis of  $\mathbb{F}^n$ : reduce the list  $u_1, \dots, u_k, e_1, \dots, e_n$  and keep the pivot columns. A *square* list with a parameter: the determinant (Example 2.9).

#### Pitfalls

- Taking the columns of the RREF as the basis: in general they do not even lie in the span.

**Practice.** Sheet 2: 2.6–2.10, 2.15, 2.19.

### 4 Sum and intersection of subspaces

Lecture 2

**Problem.** Given bases  $u_1, \dots, u_k$  of  $U$  and  $w_1, \dots, w_l$  of  $W$ , find the dimensions and bases of  $U + W$  and  $U \cap W$ .

**Method**

1. Write  $u_1, \dots, u_k$ , then  $w_1, \dots, w_l$ , into the columns; reduce.
2. The pivot columns are a basis of  $U + W$  (Section 3). Then  $\dim(U \cap W) = k + l - \dim(U + W)$  (Theorem 2.32), the number of columns without a pivot.
3. Each column without a pivot gives a relation. Put its  $u$ 's on one side and its  $w$ 's on the other: both sides are one vector, lying in  $U$  and in  $W$ . These vectors are a basis of  $U \cap W$ .

**Why step 3 works.** The  $u$ 's carry pivots, so every column without one is some  $w_j$ , and  $w_j$  occurs in no other relation: the vectors are independent, and there are  $\dim(U \cap W)$  of them (Proposition 2.24).

**Example.** Let  $u_1 = (1, 2, 0, 1)$ ,  $u_2 = (0, 1, 1, -1)$ ,  $w_1 = (1, 1, 0, 2)$ ,  $w_2 = (1, 2, -1, 1)$  and

$$U = \text{span}(u_1, u_2) \quad \text{and} \quad W = \text{span}(w_1, w_2).$$

Find the dimensions and bases of  $U + W$  and  $U \cap W$ . We write  $u_1, u_2, w_1, w_2$  into the columns of a matrix and reduce it:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 0 & 1 & 0 & -1 \\ 1 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 2 \\ 0 & \textcircled{1} & 0 & -1 \\ 0 & 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies w_2 = 2u_1 - u_2 - w_1.$$

Hence  $\dim(U + W) = 3$ ,  $\dim(U \cap W) = 2 + 2 - 3 = 1$ , and

$$U + W = \text{span}(u_1, u_2, w_1), \quad U \cap W = \text{span}(2u_1 - u_2) = \text{span}((2, 3, -1, 3)),$$

since the vector  $2u_1 - u_2 = w_1 + w_2$  lies in both  $U$  and  $W$ . ◀

**Example.** Let  $u_1 = (1, 0, 1, 2)$ ,  $u_2 = (1, 1, 0, 1)$ ,  $u_3 = (0, 1, 2, 1)$ ,  $w_1 = (2, 2, 0, 2)$ ,  $w_2 = (0, 1, 1, 0)$ ,  $w_3 = (2, 1, 0, 2)$  and

$$U = \text{span}(u_1, u_2, u_3) \quad \text{and} \quad W = \text{span}(w_1, w_2, w_3),$$

both of dimension 3. Find the dimensions and bases of  $U + W$  and  $U \cap W$ . We write the six vectors into the columns of a matrix and reduce it:

$$\begin{bmatrix} 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 2 & 1 & 1 & 2 & 0 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & 2 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & \textcircled{1} & 1 \end{bmatrix}.$$

The fourth and sixth columns give  $w_1 = 2u_2$  (so  $w_1$  already lies in  $U$ ) and  $w_3 = u_1 + u_2 - u_3 + w_2$ . Hence  $\dim(U + W) = 4$ , that is,  $U + W = \text{span}(u_1, u_2, u_3, w_2) = \mathbb{R}^4$ , and  $\dim(U \cap W) = 3 + 3 - 4 = 2$ :

$$U \cap W = \text{span}(2u_2, u_1 + u_2 - u_3) = \text{span}((2, 2, 0, 2), (2, 0, -1, 2)),$$

since  $2u_2 = w_1$  and  $u_1 + u_2 - u_3 = w_3 - w_2$  lie in both  $U$  and  $W$ . ◀

**Pitfalls**

- With the vectors in rows, as in Example 2.35, the RREF gives  $\dim(U + W)$  but no relations, and so no basis of  $U \cap W$ .

**Practice.** Sheet 2: 2.27–2.30.

## 5 Kernel and image of a matrix map

### Lecture 3

**Problem.** For  $T: \mathbb{F}^n \rightarrow \mathbb{F}^m$ ,  $Tx = Ax$ , find bases and dimensions of  $\ker T$  and  $\operatorname{im} T$ ; decide whether  $T$  is injective or surjective.

#### Method

1. Reduce the matrix  $A$ . There is no right-hand side, since it stays zero throughout.
2. *Kernel:* one vector per free variable, as for the null space in [Section 1](#);  $\dim \ker T$  is the number of free variables.
3. *Image:*  $\operatorname{im} T$  is spanned by the columns of  $A$  (Example 3.21), so reduce that list as in [Section 3](#): the pivot columns of  $A$  — not of its RREF — are a basis, and  $\operatorname{rank} T$  is the number of pivots.

**Example.** Let  $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$  be  $Tx = Ax$ . Find bases and the dimensions of  $\ker T$  and  $\operatorname{im} T$ . We reduce the matrix  $A$ :

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 2 & 3 & -5 & 4 \\ -1 & 1 & -5 & 3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 2 & -1 \\ 0 & \textcircled{1} & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies x_1 = -2x_3 + x_4, \quad x_2 = 3x_3 - 2x_4.$$

The free variables are  $x_3$  and  $x_4$ , and the pivot variables are  $x_1$  and  $x_2$ . Then:

$$\begin{aligned} x_3 = 1, \quad x_4 = 0 &\implies x = (-2, 3, 1, 0), \\ x_3 = 0, \quad x_4 = 1 &\implies x = (1, -2, 0, 1). \end{aligned}$$

Hence

$$\ker T = \operatorname{span}((-2, 3, 1, 0), (1, -2, 0, 1)), \quad \operatorname{im} T = \operatorname{span}((1, 2, -1), (2, 3, 1)).$$

Both have dimension 2:  $T$  is neither injective nor surjective. ◀

**Check.** Multiply  $A$  by each kernel vector.

#### Pitfalls

- The pivot columns of the RREF,  $(1, 0, 0)$  and  $(0, 1, 0)$ , span the plane  $x_3 = 0$ , which does not contain  $(1, 2, -1)$ : row operations change the column space.
- $4 = 2 + 2$  is no check: pivots and free variables always add up to  $n$ .

**Variant: a map from its values on a basis** (Example 3.7). Find the coordinates  $a, b, \dots$  of  $(x, y, \dots)$  in that basis by [Section 2](#), with  $(x, y, \dots)$  after the bar; then  $T(x, y, \dots) = aTv_1 + bTv_2 + \dots$ .

**Practice.** Sheet 3: 3.8, 3.10–3.12, 3.17, 3.22, 3.23, 3.29(a); 3.4 (a map from its values).

## 6 Several right-hand sides at once

### Lecture 4

**Problem.** Find the matrix of a map in given bases, convert coordinates, or find the matrix  $A' = P^{-1}AP$  of an operator in a new basis.

**Method**

1. Row operations never look at the right-hand side, so systems that share one matrix  $M$  are solved together. If  $M$  is invertible — as it is when its columns are a basis —  $[M \mid B]$  reduces to  $[I \mid M^{-1}B]$ . If the left half does not become  $I$ ,  $M$  is not invertible.
2. Choose  $M$  and  $B$  from the table below.

| Question                                  | $M$            | $B$                            | Result               |
|-------------------------------------------|----------------|--------------------------------|----------------------|
| Inverse (Section 1)                       | $A$            | $I$                            | $A^{-1}$             |
| Matrix of $T$ , basis $C$ of the codomain | $C$ as columns | $Tv_1, \dots, Tv_n$ as columns | $\mathcal{M}(T)$     |
| New coordinates                           | $P$            | $[v]_{\mathcal{B}}$            | $[v]_{\mathcal{B}'}$ |
| New matrix (Theorem 4.26)                 | $P$            | $AP$                           | $A'$                 |

All columns are coordinate vectors in one fixed basis of the codomain.  $P$  holds the new basis vectors as columns, written in the old basis (Definition 4.24). If the old basis  $\mathcal{B} = (v_1, \dots, v_n)$  is not the standard one, first find  $P$  itself: reduce  $[v_1 \cdots v_n \mid v'_1 \cdots v'_n]$  (all vectors in standard coordinates) to  $[I \mid P]$ . The other direction needs no reduction:  $[v]_{\mathcal{B}} = P[v]_{\mathcal{B}'}$ .

**Example.** An operator has the matrix  $A = \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix}$  in the standard basis. Find its matrix  $A'$  in the new basis  $(1, 1), (2, 3)$  of  $\mathbb{R}^2$ . Then  $P = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$ ,  $AP = \begin{pmatrix} 3 & 5 \\ 3 & 7 \end{pmatrix}$ , and

$$\left[ \begin{array}{cc|cc} 1 & 2 & 3 & 5 \\ 1 & 3 & 3 & 7 \end{array} \right] \xrightarrow{\text{RREF}} \left[ \begin{array}{cc|cc} \textcircled{1} & 0 & 3 & 1 \\ 0 & \textcircled{1} & 0 & 2 \end{array} \right] \Rightarrow A' = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}.$$

The right half is the matrix of the operator in the new basis. ◀

**Example (A non-standard old basis).** Let  $\mathcal{B} = ((1, 1), (1, 2))$  and  $\mathcal{B}' = ((2, 3), (1, 1))$ . A vector has  $[v]_{\mathcal{B}} = (1, 2)$ , and an operator has  $A = \mathcal{M}(T, \mathcal{B}) = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ . Find  $[v]_{\mathcal{B}'}$  and  $\mathcal{M}(T, \mathcal{B}')$ . First we find  $P$ :

$$\left[ \begin{array}{cc|cc} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{array} \right] \xrightarrow{\text{RREF}} \left[ \begin{array}{cc|cc} \textcircled{1} & 0 & 1 & 1 \\ 0 & \textcircled{1} & 1 & 0 \end{array} \right] \Rightarrow P = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then  $AP = \begin{pmatrix} 3 & 1 \\ 7 & 3 \end{pmatrix}$ , and one reduction gives both answers:

$$\left[ \begin{array}{cc|cc} 1 & 1 & 1 & 3 \\ 1 & 0 & 2 & 7 \end{array} \right] \xrightarrow{\text{RREF}} \left[ \begin{array}{cc|cc} \textcircled{1} & 0 & 2 & 7 \\ 0 & \textcircled{1} & -1 & -4 \end{array} \right].$$

So  $[v]_{\mathcal{B}'} = (2, -1)$  and  $\mathcal{M}(T, \mathcal{B}') = \begin{pmatrix} 7 & 3 \\ -4 & -2 \end{pmatrix}$ . ◀

**Check.**  $PA' = AP$ .

**Pitfalls**

- Reducing  $[P \mid A]$  gives  $P^{-1}A$ , not  $A'$ : multiply by  $P$  first.

**Practice.** Sheet 4: 4.9, 4.11, 4.12 (if the left half does not become  $I$ , find a kernel vector by Section 5); 4.16–4.21; 4.10, 4.13 (Section 3, after coordinates).

## 7 Eigenvalues and eigenspaces

### Lecture 5

**Problem.** Find the eigenvalues of a square matrix  $A$  and a basis of each eigenspace  $E(\lambda, A) = \ker(A - \lambda I)$ .

#### Method

1. Form the characteristic polynomial  $\chi_A(\lambda) = \det(\lambda I - A)$ . Expand the determinant along the row or column with the most zeros. For a  $2 \times 2$  matrix,

$$\chi_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

2. Solve  $\chi_A(\lambda) = 0$ . The roots are the eigenvalues.
3. Check: all the roots, complex ones included, add up to  $\operatorname{tr} A$  and multiply to  $\det A$  (Proposition 5.8). Count each root as many times as its multiplicity.
4. For each eigenvalue  $\lambda$ , reduce the matrix  $A - \lambda I$  and read off one vector per free variable, as for the null space in Section 1. These vectors are a basis of  $E(\lambda, A)$ .

**Example.** Find the eigenvalues of the matrix

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$$

and a basis of each eigenspace. We expand the determinant along the first row:

$$\chi_A(\lambda) = \det \begin{pmatrix} \lambda - 3 & 0 & 0 \\ -2 & \lambda - 1 & 2 \\ -2 & 2 & \lambda - 1 \end{pmatrix} = (\lambda - 3)((\lambda - 1)^2 - 4) = (\lambda - 3)^2(\lambda + 1).$$

So the eigenvalues are 3 (multiplicity 2) and  $-1$ . Check:

$$3 + 3 + (-1) = 5 = \operatorname{tr} A \quad \text{and} \quad 3 \cdot 3 \cdot (-1) = -9 = \det A.$$

For  $\lambda = 3$  we reduce the matrix  $A - 3I$ :

$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & -2 & -2 \\ 2 & -2 & -2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies x_1 = x_2 + x_3.$$

The free variables are  $x_2$  and  $x_3$ . Then:

$$\begin{aligned} x_2 = 1, x_3 = 0 &\implies x = (1, 1, 0), \\ x_2 = 0, x_3 = 1 &\implies x = (1, 0, 1). \end{aligned}$$

Hence

$$E(3, A) = \operatorname{span}((1, 1, 0), (1, 0, 1)).$$

For  $\lambda = -1$  we reduce the matrix  $A + I$ :

$$\begin{bmatrix} 4 & 0 & 0 \\ 2 & 2 & -2 \\ 2 & -2 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 \end{bmatrix} \implies x_1 = 0, \quad x_2 = x_3.$$

So the free variable is  $x_3$ . Setting  $x_3 = 1$ , we read off the basis vector  $(0, 1, 1)$  and

$$E(-1, A) = \operatorname{span}((0, 1, 1)).$$

The double eigenvalue 3 has a two-dimensional eigenspace.

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**Check.** Verify  $Av = \lambda v$  for each vector found. If the RREF of  $A - \lambda I$  has no free variable,  $\lambda$  is not an eigenvalue: look for the error in  $\chi_A$ , in its roots or in the reduction.

### Pitfalls

- ▶ Eigenvalues cannot be read off the diagonal of an RREF: row operations change the eigenvalues. The diagonal holds the eigenvalues only when  $A$  itself is triangular.
- ▶ A double root does not guarantee two independent eigenvectors:  $\dim E(\lambda, A)$  is the number of free variables and can be smaller than the multiplicity (Example 5.11).

**Variants.** A *triangular* matrix: the eigenvalues are its diagonal entries (Theorem 5.21, Example 5.10), so steps 1–2 are not needed. A *real* matrix, working over  $\mathbb{C}$ : for a non-real eigenvalue  $\lambda$ , reduce  $A - \lambda I$  only. The conjugates of the basis vectors found are a basis of  $E(\bar{\lambda}, A)$ , since  $A\bar{v} = \overline{Av} = \bar{\lambda}\bar{v}$ . An *operator* on another space: take its matrix in some basis (for polynomials, the standard one). The vectors found are the coordinate vectors of the eigenvectors: turn them back into vectors of that space.

**Practice.** Sheet 5: 5.1, 5.4, 5.5, 5.10; 5.2, 5.6 (over  $\mathbb{C}$ ); 5.3 (a space of polynomials); 5.17 (a triangular matrix).